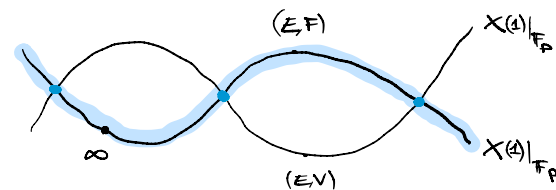
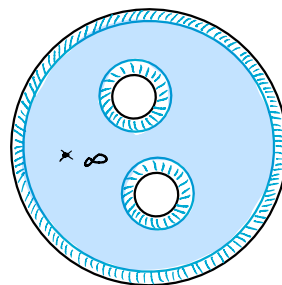
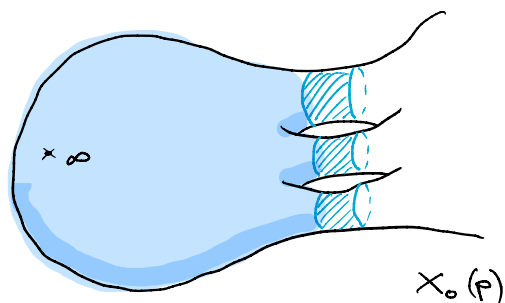


# P-ADIC MODULAR FORMS & THEIR GEOMETRY

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LEIDEN UNIVERSITY



JULY 23 2025  
SYMPAAN' 25

# § I. INTRODUCTION:

1

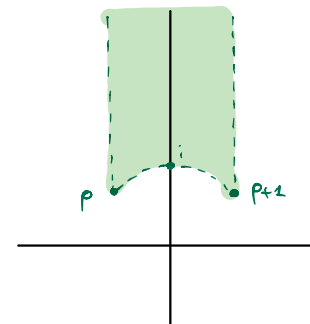
## THREE PERSPECTIVES ON MODULAR FORMS

### ■ COMPLEX ANALYTIC

$f: \mathcal{H} \longrightarrow \mathbb{C}$  HOLOMORPHIC,  $k \in \mathbb{Z}$

$$f\left(\frac{az+b}{cz+d}\right) = (cz+d)^{-k} f(z), \quad z \in \mathcal{H}, \quad \begin{matrix} \text{FINITE INDEX} \\ \uparrow \\ (a \ b) \\ (c \ d) \end{matrix} \in \Gamma \subseteq SL_2(\mathbb{Z})$$

+ EXTRA CONDITIONS



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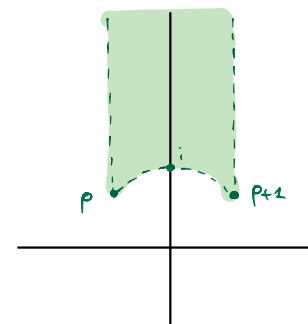
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FINITE INDEX

† EXTRA CONDITIONS



### ■ GENERATING SERIES

$$(q = e^{2\pi i z})$$

$$- G_k(q) = -\frac{B_k}{2k} + \sum_{n \geq 1} \sigma_{k-1}(n) q^n, \quad \sigma_{k-1}(n) = \sum_{d \mid n} d^{k-1}, \quad B_k = -k \cdot \sum_{n \geq 1} (1-n)$$

EISENSTEIN SERIES

NUMBER THEORETIC FLAVOUR

$$- \eta(q)^{-1} = q^{-\frac{1}{24}} \prod_{n \geq 1} (1-q^n)^{-1} = q^{-\frac{1}{24}} \sum_{n \geq 0} P(n) q^n, \quad P(n) = \# \{ \text{PARTITIONS OF } n \}$$

DEDEKIND  $\eta$ -FUNCTION

COMBINATORIAL FLAVOUR

$$- f \in S_2(\Gamma_0(N), \mathbb{Q}) \quad f(q) = \sum_{n \geq 1} a_n \cdot q^n, \quad a_p = p+1 - \# E_f(\mathbb{F}_p) \quad p \nmid N$$

WEIGHT 2 EIGENFORM

GEOMETRIC FLAVOUR

$E_f/\mathbb{Q}$  ELLIPTIC CURVE ATTACHED TO  $f$

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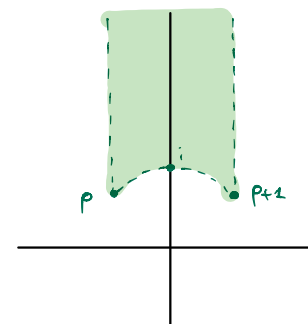
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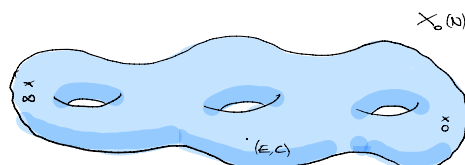
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$E_f/\mathbb{Q}$  ELLIPTIC CURVE ATTACHED TO  $f$

### GEOMETRIC

$$f \in H^0(X_0(N), \omega^*)$$

SECTION OF LINE BUNDLE



$E$  ELLIPTIC CURVE

$C \in E[N]$  CYCLIC  $N$ -GROUP

MODULI SPACE OF ELLIPTIC CURVES

W/ CYCLIC  $N$ -SUBGROUP



# CONGRUENCES BETWEEN MODULAR FORMS

2

- $\Delta(q) = \eta^{24}(q) = q \cdot \prod_{n \geq 1} (1 - q^n)^{24} = \sum_{n \geq 1} \tau(n) q^n$  DISCRIMINANT MODULAR FORM
- $G_{12}(q) = \frac{691}{65520} + \sum_{n \geq 1} \sigma_{11}(n) q^n$  EISENSTEIN SERIES

RAMANUJAN'S CONGRUENCE ('16)

$$\tau(p) \equiv 1 + p^{11} \pmod{691}.$$

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## SERRE'S IDEA ('72):

CONSTRUCT P-ADIC ANALYTIC FAMILIES OF MODULAR FORMS FROM THEIR q-EXPANSION

$$f(q) = a_0 + a_1 q + a_2 q^2 + \dots \in \mathbb{Q}_p[[q]], \quad v_p(f) := \inf_n (v_p(a_n))$$

THE SPACE OF P-ADIC MODULAR FORMS IS COLLECTION OF  $f(q) \in \mathbb{Q}_p[[q]]$  SUCH THAT

$$v_p(f(q) - f_i(q)) \longrightarrow \infty \quad \text{FOR } f_i \in M_{k_i}(SL_2(\mathbb{Z}), \mathbb{Q}) \text{ CLASSICAL MODULAR FORMS}$$

# P-ADIC FAMILY OF EISENSTEIN SERIES $(p \geq 5)$

3

KOHMER CONGRUENCES ('51)

LET  $K \equiv K' \pmod{(p-1)p^n}$

$$\text{IF } (p-1) \nmid K : (1-p^{K-1})B_K/K \equiv (1-p^{K'-1})B_{K'}/K' \pmod{p^{n+2}}$$

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## KUMMER CONGRUENCES ('51)

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CONSIDER  $G_K^{(p)}(q) := (1-p^{K-1}) \frac{-B_K}{2K} + \sum_{n \geq 1} \left( \sum_{p \nmid d|n} d^{K-1} \right) q^n$  MODULAR FORM FOR  $\Gamma_0(p)$

$\leadsto$  BY FERMAT'S LITTLE THEOREM AND KUMMER CONGRUENCES FOR  $K \equiv K' \pmod{(p-1)p^n}$

$$\bullet \text{ IF } (p-1) \nmid K : G_K^{(p)}(q) \equiv G_{K'}^{(p)}(q) \pmod{p^{n+1}}$$

LET  $K \in \mathbb{Z}_p \times \mathbb{Z}/(p-1)\mathbb{Z}$ ,  $\sigma_{K-1}^*(n) = \sum_{p \nmid d|n} d^{K-1}$  P-ADIC INTEGER

$$G_K^{(p)} := \lim_{K_i} G_{K_i}^{(p)} = \frac{1}{2} \sum^* (1-K) + \sum_{n \geq 1} \sigma_{K-1}^*(n) q^n \quad \text{IS A P-ADIC MODULAR FORM}$$

$$\sum^* (1-K) = \lim_{i \rightarrow \infty} \sum (1-K_i) \quad K_i \rightarrow K, \quad K_i \text{ INTEGERS}$$

$\hat{=}$   
KUBOTA-LEOPOLD P-ADIC  $\sum$ -FUNCTION

# TOWARDS A FINER DESCRIPTION

4

## ■ HECKE STRUCTURE

(CLASSICAL) MODULAR FORMS OF LEVEL  $N$  ARE EQUIPPED W/ ACTION OF HECKE ALGEBRA

$$\begin{aligned} \bullet T_\ell f(q) &= \sum_{n \geq 0} a_{\ell n} q^n + \ell^{k-1} \sum_{n \geq 0} a_n q^{\ell n}, & \ell \nmid N \\ \bullet U_p f(q) &= \sum_{n \geq 0} a_{np} q^n, & p \mid N \end{aligned}$$

$U_p$  OPERATOR ON  $p$ -ADIC MODULAR FORMS HAS LARGE CONTINUOUS SPECTRUM

$$\lambda \in p\mathbb{Z}_p, \quad f_\lambda := (1 - \lambda V_p)^{-1} (1 - V_p U_p) f$$

$$U_p f_\lambda = \lambda f_\lambda$$

**PROBLEM:** WE CANNOT DECOMPOSE  $p$ -ADIC MF IN FINITE LINEAR COMBINATION EIGENFORMS!

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## **KATZ'S IDEA** ('72)

INTRODUCE  **$p$ -ADIC GEOMETRIC FRAMEWORK:**

**OVERCONVERGENT MODULAR FORMS**

# § II. GEOMETRY OF MODULAR CURVES

(p>5)

5

## COMPLEX ANALYTIC VS SCHEMES

MODULI SPACE OF ELLIPTIC CURVES WITH CYCLIC  $p$ -SUBGROUP

### ■ COMPLEX PICTURE

$$\Gamma_0(p) = \left\{ \gamma \in SL_2 \mathbb{Z} : \gamma \equiv \begin{pmatrix} * & * \\ 0 & * \end{pmatrix} \pmod{p} \right\}, \quad Y_0(p) = \mathbb{P}^1 \setminus \mathcal{H} \longrightarrow \{ \text{ELLIPTIC CURVES} / \mathbb{C} \text{ w/ } p\text{-SUBGRP} \}$$
$$\simeq \longmapsto (\mathbb{C} / (\mathbb{Z} + \tau \mathbb{Z}), \frac{1}{p})$$

$X_0(p)$  ITS COMPACTIFICATION IS RIEMANN SURFACE

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### ■ SCHEME THEORETIC PICTURE

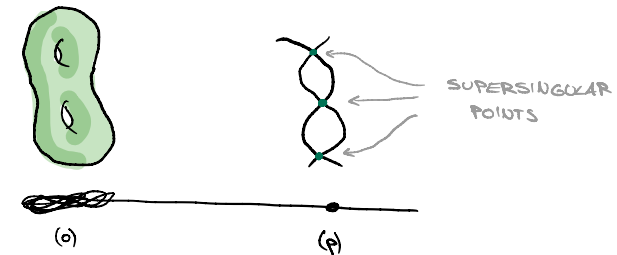
Rings  $\longrightarrow$  Sets

$$R \longmapsto \left\{ E/R \text{ ELLIPTIC CURVE} \right. \\ \left. \text{w/ } p\text{-SUBGROUP} \right\} / \sim$$

REPRESENTED BY SMOOTH SCHEME  $Y_0(p)$  OVER  $\mathbb{Z}[1/p]$

TAKE  $X_0(p)$  ITS COMPACTIFICATION, SMOOTH PROPER OVER  $\mathbb{Z}[1/p]$

DELIGNE - RAPAPORT MODEL: SEMISTABLE MODEL OVER  $\mathbb{Z}$





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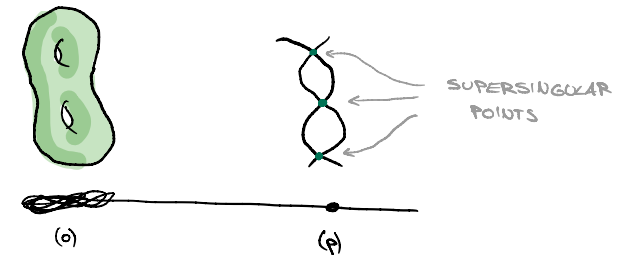
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CONNECTION WITH MODULAR FORMS:  $L/\mathbb{Q}$  FIELD EXTENSION,  $k \in \mathbb{Z}$

$$S_{2k}(\Gamma_0(p), L) = H^0(X_0(p)_L, \Omega_{X_0(p)/L}^{\otimes k}) \quad \text{CUSP FORMS OF WEIGHT } 2k$$

REM. TAKING MODULAR CURVE OVER  $\mathbb{C}_p$  WILL NOT ADD ANY NEW MODULAR FORM  $\implies$  NEED RIGID ANALYTIC SETTING

# RIGID PICTURE: ORDINARY LOCUS & SUPERSINGULAR ADDICI

6

CONSIDER  $E/\mathbb{F}_q$  ELLIPTIC CURVE OVER FINITE FIELD

$$E[p](\mathbb{F}_q) \simeq \mathbb{Z}/p\mathbb{Z}$$

ORDINARY EC

$$E[p](\mathbb{F}_q) = \{0\}$$

SUPERSINGULAR EC

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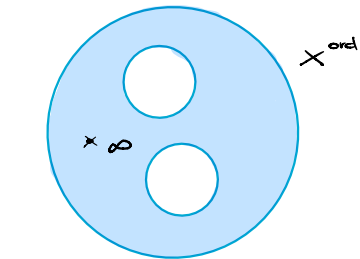
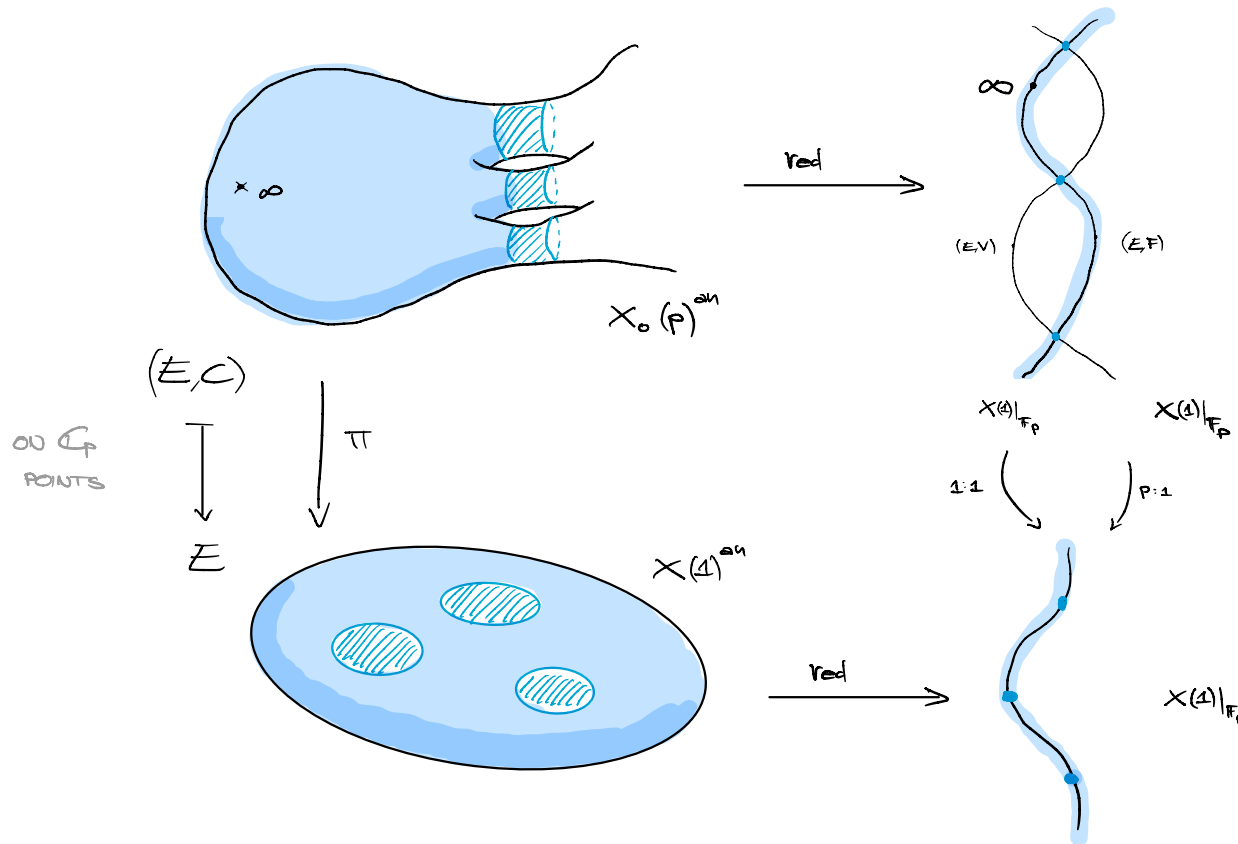
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RIGID ANALYTIFICATION OF MODULAR CURVES



  $\rightarrow$  ORDINARY LOCUS

  $\rightarrow$  SUPERSINGULAR LOCUS

**ACHTUNG:** A PROPER TREATMENT SHOULD INVOLVE EXTRA STRUCTURE TO AVOID PROBLEMS OF REPRESENTABILITY OF MODULI PROBLEMS.

# RIGID PICTURE: ORDINARY LOCUS & SUPERSINGULAR AFFOID

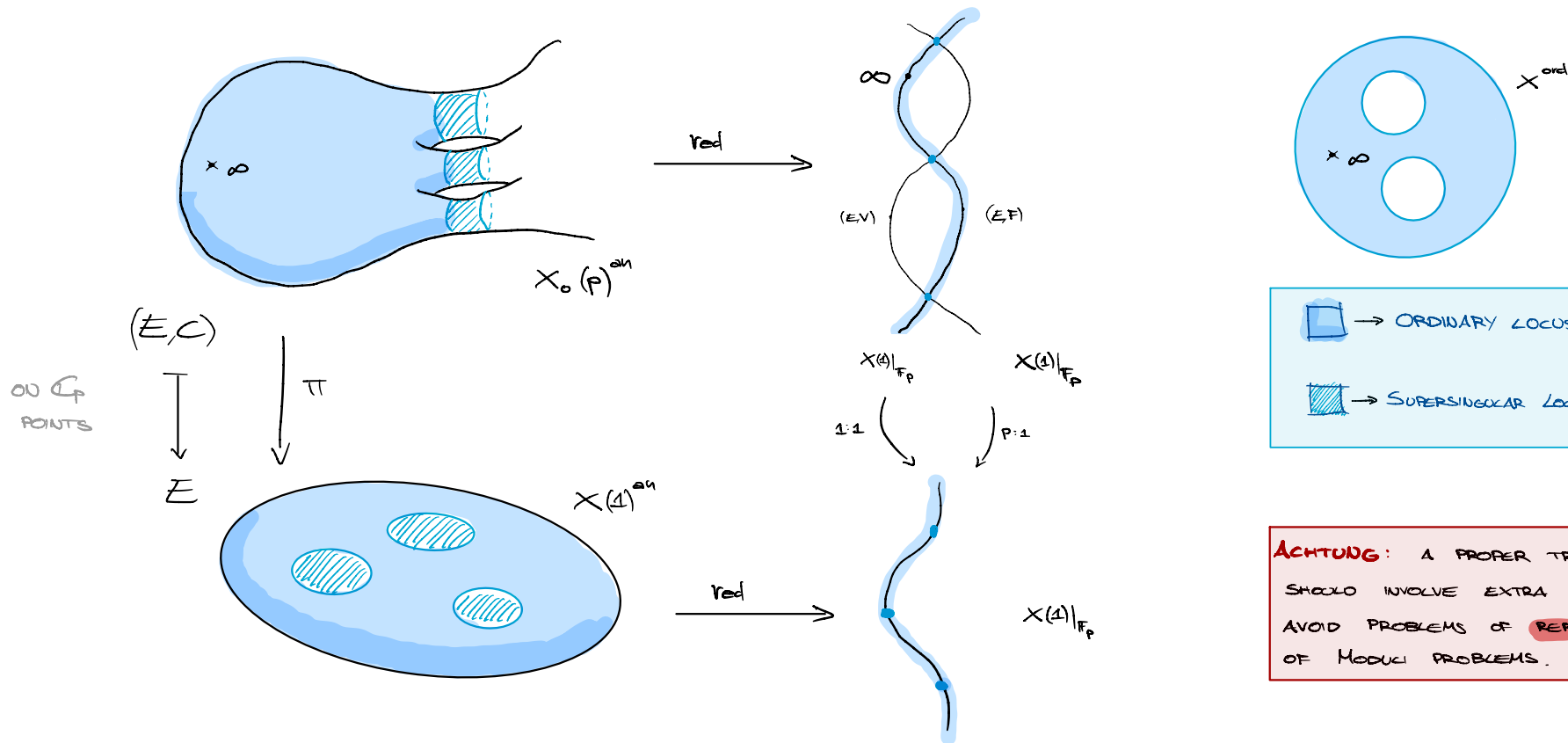
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RIGID ANALYTIFICATION OF MODULAR CURVES



THE AFFINOID  $X^{\text{ord}}$  IS CALLED **ORDINARY LOCUS** AND CORRESPONDS TO ELLIPTIC CURVES WITH ORDINARY REDUCTION

**REM**  $p$ -ADIC MODULAR FORMS à LA SERRE (OF CLASSICAL WEIGHTS) CORRESPOND TO SECTIONS OVER ORDINARY LOCUS

$$S_{2k}^{(p)}(\mathcal{L}) := \Omega_{\text{rig}}^{\text{ex}}(X_{\mathcal{L}}^{\text{ord}})$$

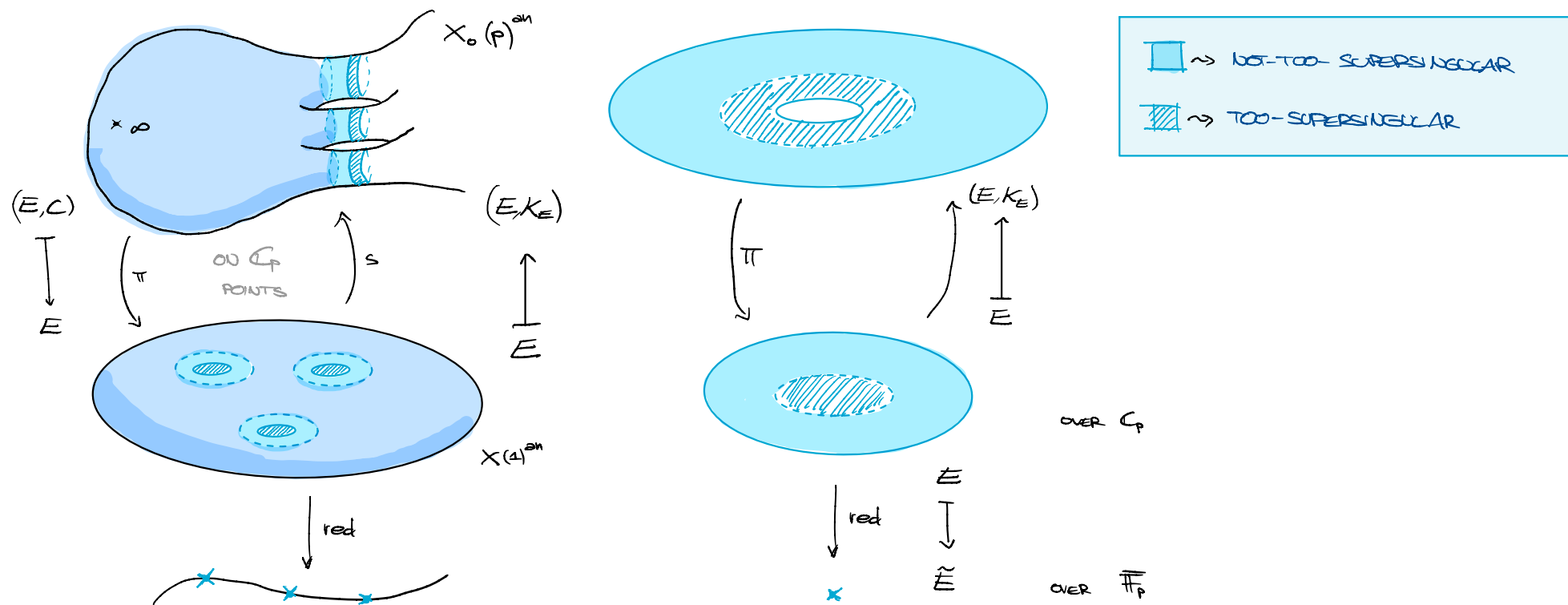
$\mathcal{L}/\mathbb{Q}_p$  FIELD EXTENSION

# CANONICAL SUBGROUP: LUBIN'S IDEA

7

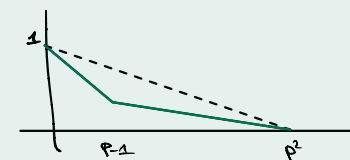
SUPERSINGULAR ANNULI HAVE SPECIAL LOCUS  $\leadsto$  NOT TOO SUPERSINGULAR LOCUS

ASSOCIATE TO ELLIPTIC CURVE A CANONICAL SUBGROUP



WHERE DOES CANONICAL SUBGROUP COME FROM?

- ORDINARY LOCUS:  $K_E = \text{Ker}(E[p] \rightarrow E[p](\mathbb{F}_p))$
- SUPERSINGULAR LOCUS: NEWTON POLYGON OF  $[p]$  ON  $\hat{E}$  FORMAL GROUP



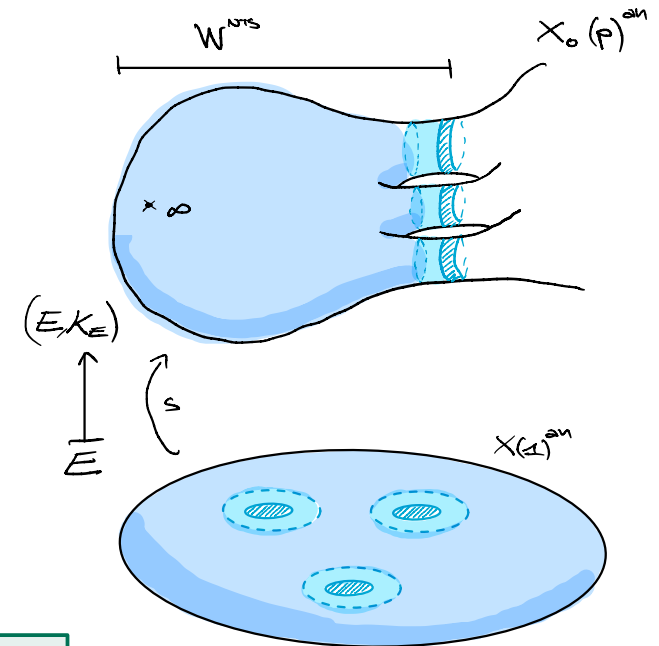
# OVERCONVERGENT MODULAR FORMS

8

CONSIDER  $W^{NTS}$  THE ORDINARY LOCUS TOGETHER WITH NOT-TOO-SUPERSINGULAR LOCUS

$$X^{ord} \subset W^{NTS}$$

$W^{NTS} \setminus X^{ord}$  FINITE DISJOINT UNION OF OPEN ANNULI



AN OVERCONVERGENT P-ADIC MODULAR FORM IS A P-ADIC MODULAR FORM THAT IS DEFINED ON  $W^{NTS}$  (OVERCONVERGES IN PART OF SUPERSINGULAR LOCUS)

$$S_{2k}^+(L) = \Omega_{rig}^{\otimes k}(W_L^{NTS}), \quad L/\mathbb{Q}_p \text{ FIELD EXTENSION.}$$

REM CLASSICAL MODULAR FORMS OF LEVEL  $\Gamma_0(p)$  NATURALLY SIT IN OVERCONVERGENT P-ADIC MODULAR FORMS

# $U_p$ OPERATOR : THE REASON EVERYTHING WORKS

9

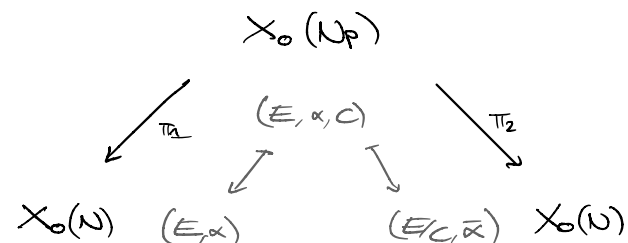
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GIVEN BY GEOMETRIC CORRESPONDENCE

$\sim$  PRESERVES ORDINARY LOCUS

NOT-TOO-SS LOCUS



THEOREM  $U_p$  IS A CONTINUOUS BOUNDED COMPACT OPERATOR ON  $S_{2k}^+$   
I.E. IMAGE OF UNIT BALL IS RELATIVELY COMPACT

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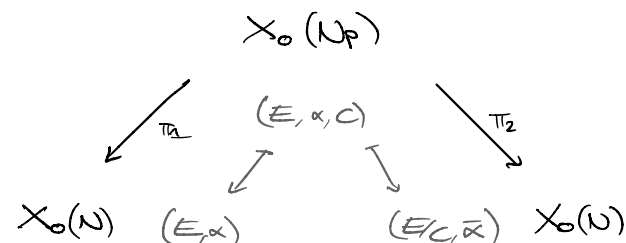
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SERRE AND DWORK ('62) PROVED FOLLOWING PROPERTIES OF THESE OPERATORS

■  $U_p$  IS LIMIT OF OPERATORS OF FINITE RANK

■  $U_p$  HAS WELL DEFINED TRACE

■  $U_p$  ADMITS A DISCRETE SPECTRUM

$$|\lambda_1| \geq |\lambda_2| \geq \dots \geq \dots \quad \text{w/} \quad |\lambda_i| \rightarrow 0$$

AND SEQUENCE OF GENERALIZED EIGENFORMS  $f_i$  SUCH THAT EVERY OVERCONVERGENT MF  $f$  ADMITS

$$f \sim \sum \alpha_i f_i \quad \text{ASYMPTOTIC EXPANSION} \quad \left\| U^n f - \sum_{i \in I} \alpha_i U^n f_i \right\| = o(\varepsilon^n)$$



# § III. COMPUTING $U_p$ -EIGENFUNCTIONS

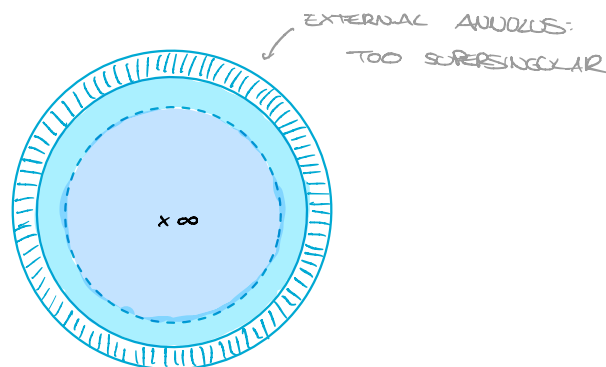
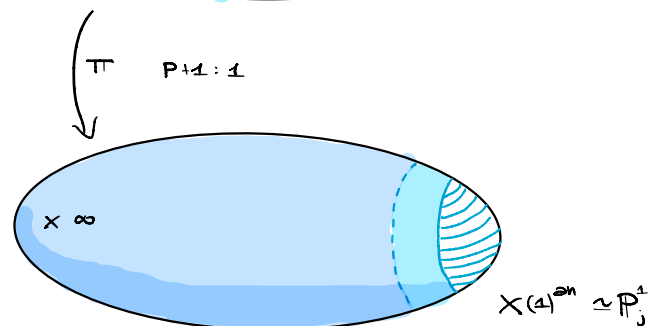
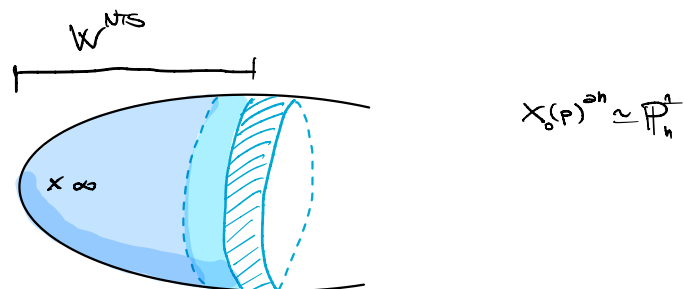
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## GENUS 0 CASE

LET  $p$  BE 2, 3, 5, 7 OR 13.

$$h_p(z) = \left( \frac{\Delta(pz)}{\Delta(z)} \right)^{\frac{1}{p-1}}$$

HAUPTMODUL IS UNIFORMIZER FOR  $X_0(p)^{\text{an}}$



$$W^{\text{uts}}(\mathbb{C}_p) \simeq \left\{ z \in \mathbb{C}_p \cup \{\infty\} \mid |h_p(z)| < p^{\frac{12 \cdot p}{p^2-1}} \right\} \simeq B(0, p^{\frac{12 \cdot p}{p^2-1}})$$

$$X^{\text{ord}}(\mathbb{C}_p) \simeq \left\{ z \in \mathbb{C}_p \cup \{\infty\} \mid |h_p(z)| \leq 1 \right\} \simeq B(0, 1]$$

TWO RIGID  $p$ -ADIC DISCS CENTERED AT THE CUSP  $\infty$

EG  $p=2, j = \frac{(1+2^8 \cdot h_2)^3}{h}$

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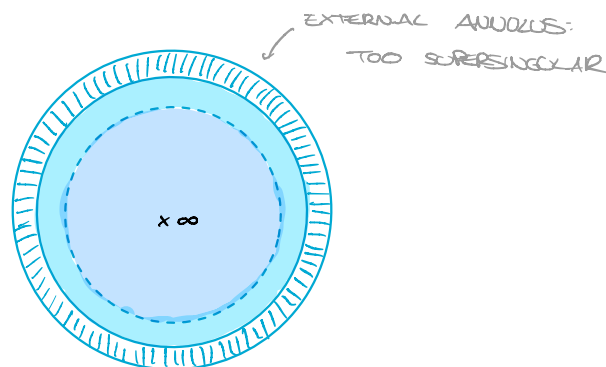
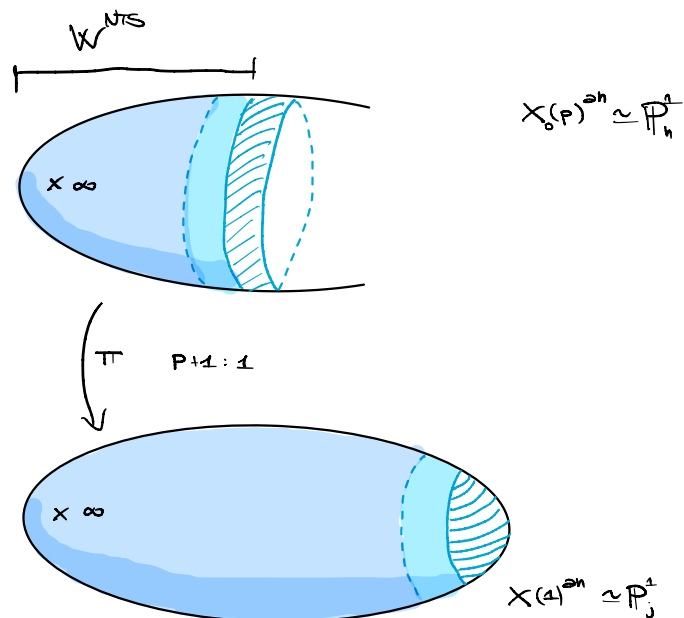
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LET  $p$  BE 2, 3, 5, 7 OR 13.

$$h_p(z) = \left( \frac{\Delta(pz)}{\Delta(z)} \right)^{\frac{1}{p-1}}$$

HAUPTMODUL IS UNIFORMIZER FOR  $X_0(p)^{\text{an}}$



- $W^{\text{UTS}}(\mathbb{C}_p) \simeq \{ z \in \mathbb{C}_p \cup \{\infty\} \mid |h_p(z)| < p^{\frac{12 \cdot p}{p^2-1}} \} \simeq B(0, p^{\frac{12 \cdot p}{p^2-1}})$
- $X^{\text{ord}}(\mathbb{C}_p) \simeq \{ z \in \mathbb{C}_p \cup \{\infty\} \mid |h_p(z)| \leq 1 \} \simeq B(0, 1]$

EG  $p=2, \quad j = \frac{(1+2^8 \cdot h_2)^3}{h}$

TWO RIGID **P-ADIC DISCS** CENTERED AT THE CUSP  $\infty$

THE WEIGHT 0 OVERCONVERGENT  $p$ -ADIC MODULAR FORMS CORRESPONDS TO FUNCTION ON  $p$ -ADIC DISK  $W^{\text{UTS}}$

I.E. LAURENT SERIES  $f = a_0 + a_1 p^r h_p + a_2 p^{2r} h_p^2 + \dots \in \mathbb{C}_p[[p^r h_p]]$ ,  $r = \frac{12p}{p^2-1}$  WITH GROWING CONDITIONS ON COEFFICIENTS

# $U_p$ ACTION ON HAUPTMODUL

11

FROM THE GEOMETRIC DESCRIPTION  $\leadsto$  OVERCONVERGENT MODULAR FUNCTIONS EXPRESSED AS  
POWER SERIES ON  $h_p$  (+ GROWING CONDITIONS)

$U_p$  ACTION ON  $S_0^+(\Gamma_p)$  GIVEN BY  $U_p$  ACTION ON  $p^r h, p^{2r} h^2, p^{3r} h^3, \dots$   $r = \frac{12 \cdot p}{p^2 - 1}$

PRO USE GENUS 0 OF  $X_0(p)$  + DESCRIPTION OF  $U_p$  AS CORRESPONDENCE TO OBTAIN

$$U_p h^i = P_i(h)$$

$P_i(x) \in \mathbb{C}_p[x]$  POLYNOMIAL  $\leadsto$  COMPUTATIONALLY PRECISE

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EG  $p=2$

$$V_2(U_2(i,j))_{ij} = \begin{pmatrix} 3 & 8 \\ 3 & 7 & 11 & 16 \\ 8 & 12 & 17 & 19 & 24 \\ 7 & 11 & 15 & 21 & 23 & 27 & 32 \\ 11 & 19 & 20 & 25 & 27 & 35 & 35 & \dots \\ 11 & 16 & 20 & 24 & 27 & 33 & 35 \\ 17 & 19 & 24 & 29 & 34 & 35 \\ 19 & 20 & 23 & 27 & 31 & 38 \\ 19 & 24 & 27 & 37 & 36 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

$$U_2(h^n) = (48h + 4096h^2) U_p(h^{n-1}) + h U_p(h^{n-2})$$

THEOREM (BUZZARD - CALLEGARI, '05)

THE SLOPE SEQUENCE OF  $U_2$  ON  $S_2^+(\mathbb{C}_p)$   
IS GIVEN BY

$$\left\{ 1 + 2v_2\left(\frac{(3n)!}{n!}\right) \right\}_{n=1, \dots}$$

# A MYSTERIOUS BEHAVIOUR

12

CONSIDER  $p=2$ , LET  $f_i$  EIGENFORMS FOR  $U_2$ -OPERATOR,  $U_2 f_i = \lambda_i f_i$   $\lambda_i \in \mathbb{C}_2$

$$V_2(\lambda_1) \leq V_2(\lambda_2) \leq V_2(\lambda_3) \leq \dots$$

CALLEGARI ('13) OBSERVED THAT

$q$ -EXPANSION OF  $f_{2^n}$  APPEAR TO CONVERGE TO AN INFINITE SLOPE FORM

$$E_0^{[2]}(q) = \sum_{n \geq 1} \left( \sum_{2 \nmid d|n} d^{-2} \right) q^n$$

COMING FROM INTEGRAL OF  $p$ -DEPLETION OF A CLASSICAL MODULAR FORM

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50 YEARS AFTER SERRE'S FIRST RESULTS, LIMITS OF  $q$ -EXPANSIONS OF  $p$ -ADIC MODULAR FORMS REMAIN RICH AND NOT WIDELY UNDERSTOOD